

Transitioning to Sustainable Aviation Fuels: Case of the Philippine Aviation Industry

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Abstract

The low energy and power density of batteries provides a challenge in transitioning the aviation industry from fossil fuel-based energy carriers. To overcome this issue, sustainable aviation fuels (SAF) such as HEFA, alcohol-to-kerosene, and e-fuels are proposed to be viable alternatives to conventional jet fuel. This paper investigates the energy, land, and investment needs to transition from jet fuel to SAF. The Philippines was used as a case study for its abundance in renewable energy resources. For the technology, the production of e-kerosene using carbon dioxide from Direct Air Capture (DAC) technology powered by solar or wind energy was considered. To ensure a successful transition using this pathway, technology prices should drop down in the coming decades. Moreover, the huge land footprint for the needed renewable energy plants could be an issue if not managed. Investment of 7.9 billion euros was calculated for transitioning the current jet fuel demand to SAF for the Philippines.

1. Introduction

The continued dependence on fossil fuels has caused anthropogenic greenhouse gases to surpass global thresholds. This has raised urgency in finding a new sustainable source of energy through renewable alternatives. However, the transition to cleaner energies has its limitations. In land transportation, the use of diesel and gasoline in cars, buses, and trucks may be replaced by electric vehicle technology. On the other hand, in air transportation, the use of energy stored in batteries is not feasible and is unlikely to change in the foreseeable future. Low power density and range issues in battery technology have been identified to cause this problem. Thus, there is a need to find innovative ways for energy storage to power the shift of the current aviation industry to a cleaner direction.

Sustainable aviation fuels (SAF) offer a potential solution in decarbonizing the air transport industry by generating fuels with identical performance of current jet fuels yet are not produced from fossil fuels. In this case, the energy density would be the same as regular fuels but with near zero carbon emissions during production ([Rojas-Michaga, 2023](#)). However, minimal emissions from this technology would only be achieved if the energy requirement for SAF production is sourced from renewable energy. Moreover, the production of SAF is a highly energy intensive process which would require investments and a large land footprint for the necessary renewable energy power plants. Therefore, understanding the technical and financial requirements for the transition to SAF would be necessary to plan forward.

In this paper, the following steps were conducted:

- the current state of technology for producing SAF are explored,
- energy requirements for production of SAF for the study site was computed,
- land footprint for the additional renewable energy capacity was determined, and
- investments for the transition were estimated.

In this case, the study area selected was the Philippines - an archipelagic nation that relies on the local aviation industry for travel within its islands. The country is also geographically located in areas with high renewable energy potential especially for solar and wind projects due to its position near the equator and along the strong winds from the Pacific Ocean.

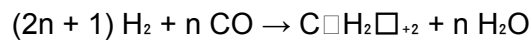
2. Literature Review

2.1 E-fuels

E-fuels, also known as Power-to-Liquid fuels (PtL) or PtX liquids, are synthesized hydrocarbons similar to traditional fossil-fuel derived fuels like diesel, gasoline, and kerosene. The process of generating PtL fuels is electricity-intensive and currently uneconomical. Technically, the CO₂, H₂, and electricity used for the production can be from the fossil fuel industry and other high-carbon emitting industries. However, to make the technology sustainable, pairing with renewable and carbon capture technologies is necessary. Common PtL fuels are e-diesel, e-gasoline, and e-kerosene which are the same as their fossil fuel counterparts but differ in their process of production ([Lilley, n.d.](#)).

2.2 Fischer-Tropsch Synthesis

The Fischer-Tropsch process is a catalytic chemical reaction that converts a mixture of CO and hydrogen H₂ into liquid hydrocarbons such as synthetic diesel, kerosene, gasoline, and waxes as shown in the formula below:



2.3 Sustainable Aviation Fuels

Currently, high-grade kerosene is conventionally used for jet fuel. Meanwhile, SAF is a potential candidate to replace traditional fossil-fuel derived jet fuels. There are multiple technological routes to produce SAFs such as Hydroprocessed Esters and Fatty Acids (HEFA), Alcohol-to-Jet, and PtL fuels like e-kerosene, enabling flexibility in different regions based on available resources and renewable energy. Currently, HEFAs are leading in commercial maturity but e-kerosene is expected to be commercially viable in the future with more added benefits. For this study, the SAF transition will be focused on e-kerosene ([Grim, 2022](#)).

2.4 Direct Air Capture

Direct air capture (DAC) is a technology that pulls CO₂ directly from the atmosphere through membrane technology that can then be either stored underground or used to create products such as e-fuels. In this case, DAC was assumed to be the CO₂ source of the PtL technology. The technology is currently still in pilot-scale.

3. Methodology

3.1 Study Area

The Philippines is located in Southeast Asia on the west side of the Pacific and near the equator. Because of this, the country has access to a lot of renewable energy resources. The country's aviation sector fuel consumption is around 33.7 thousand barrels of jet fuel per day. This will be the basis for the energy demand used in this work. Table 1 summarizes the important data from the Philippines used in this paper.

Table 1. Fuel demand and renewable energy potential and installed capacity

Parameters	Value	Unit
Amount of Fuel for Aviation Industry PH	33,700.00	barrels per day
Commercial Solar Potential	31.00	GW
Solar Installed Capacity	1,443.00	MW
Commercial Wind Potential	89.00	GW
Wind Installed Capacity	443.00	MW

3.2 Energy Requirements

Data for the energy requirements of this study were gathered from multiple sources, the main energy conversion efficiency of power to fuel was based on a DAC-sourced CO₂ as seen in Figure 1 ([Seymour et al. 2024](#)). This was done to highlight how much energy, land, and financial resources would be necessary to power the study area's aviation sector fully by DAC-based e-kerosene. For electricity, only solar and wind resources were considered due to available data. Table 2 shows the energy conversion data.

Table 2. Fuel production energy requirement data

Parameters	Value	Unit
Energy to produce e-kerosene from DAC	10.00	kWh per liter
Energy to produce efuel blend	25.20	kWh per liter
Jet fuel fraction of e-fuel blend	0.40	l e-kerosene per l efuel

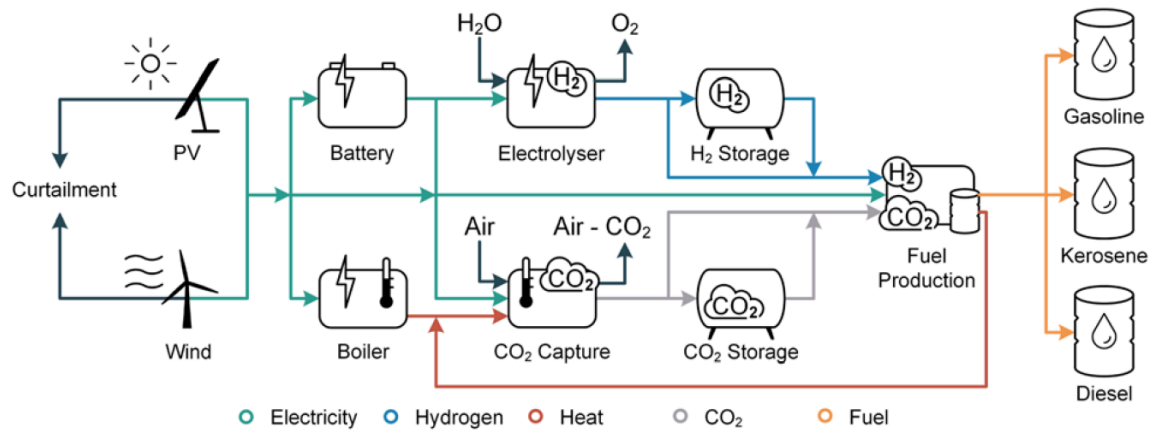


Figure 1. Configuration of e-fuel production plant assumed in this study

3.3 Land Requirement

Land footprint of the entire transition of air industry to e-kerosene was calculated by values coming from sources such as National Renewable Energy Laboratories (NREL) and other available papers (Bolinger, 2022). Table 3 shows the estimated land footprint technologies for both solar and wind energies.

Table 3. Fuel production energy requirement data

Parameters	Value	Unit
Solar Power Land Footprint	1.15	ha/MW
Wind Power Land Footprint	34.00	ha/MW

3.4 Investments

Because the technology is still in a pilot-scale development phase, limited information can be gathered. However, some papers have estimated the cost for development of these projects at large scale and were used to calculate the estimated required RE investment to transition to a SAF-fueled aviation sector (Sens, 2022). To compare, CAPEX estimates for 2030 and 2050 scenarios were used. Table 4 shows the economic data.

Table 4. CAPEX and OPEX projections 2030 and 2050

Parameter	2030	2050	Unit
Solar PV CAPEX	430	330	eur per kW
Wind Onshore CAPEX	1210	970	eur per kW
E-fuel Plant CAPEX	0.85	0.45	eur per l

4. Results

The calculated jet fuel consumption of the country is 165,130 liters per hour. This requires an additional installed capacity of 8,322.55 MW of solar PV or 5,548.37 MW of wind. The difference in energy requirement from different sources is attributed to the capacity factor of each technology: 0.2 for solar and 0.3 for wind. The amount of additional solar energy needed for SAF production is almost equal to the currently installed RE capacity in the country. The country intends to increase renewable energy share of its grid by 50% in 2040 by installing 20 GW of renewable energy. This would mean that an additional 28 to 42% of the country's 2040 goal will be required to supply the country's jet fuel needs by DAC as seen on Figure 2.

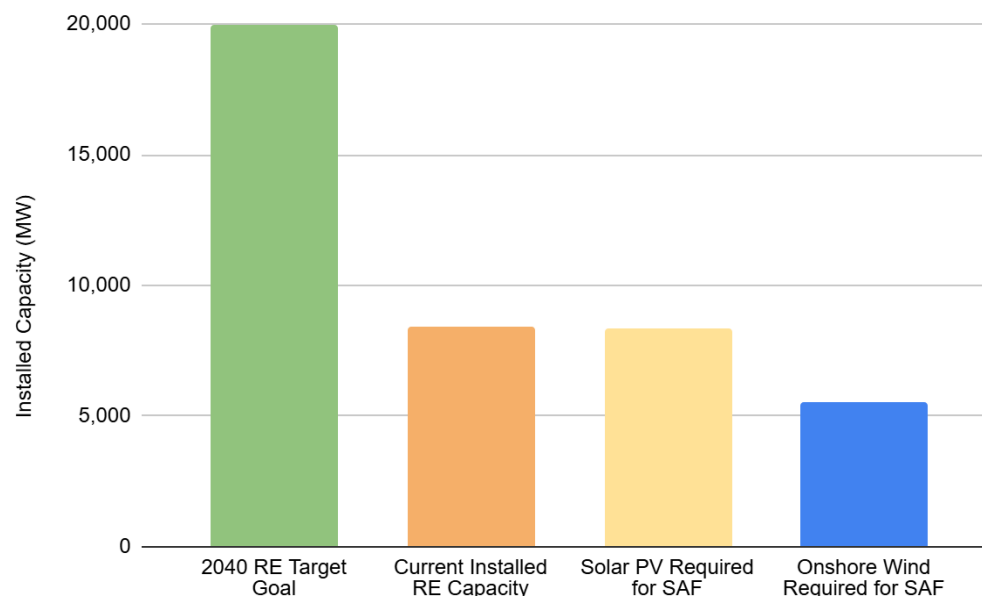


Figure 2. Additional Solar and Wind Installed Capacities for SAF compared to the country's 2040 RE Goals

To support the installation of 8.3 GW of solar energy, 9,566.15 hectares of land would be required. This would be enough for 18 to 23 million solar panels installed in areas that have medium to high solar energy potentials in the country. On the other hand, onshore wind would need 1,850 to 2,775 wind turbines spread across a land area of 188,644.51 hectares. The land footprint of solar energy is only 5% of wind energy. This is caused by the required distance in between turbines. This shows an interesting dynamic between land footprint and installed capacity of both renewable sources. Figure 3 shows a visualization of the land footprints in one of the areas with sufficient renewable energy potential.



Figure 3. Comparison of land footprint for solar energy (yellow) and wind energy (red) requirements

The total CAPEX for the transition is around 4.8 to 7.9 billion euros (2025) for technology prices in 2030. Because of the foreseen decrease in renewable energy and DAC in the next decade, the CAPEX decreases to 3.4 to 6 billion euros by 2050. OPEX is expected to be around 20% in all cases as seen in Figure 4.

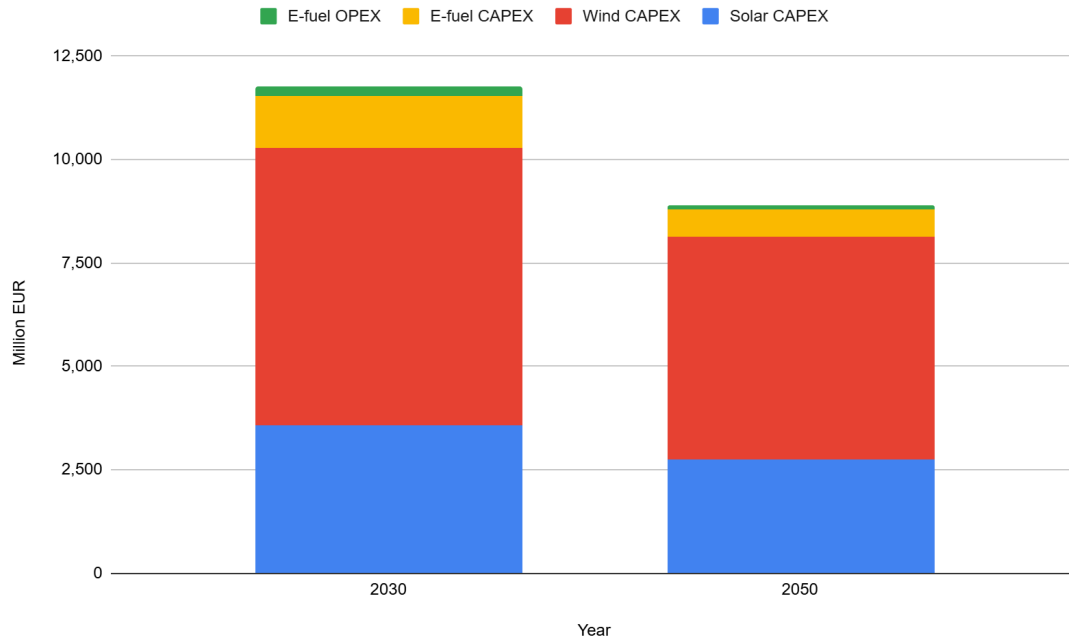


Figure 4. Investment needed for SAF transition with projected price of technologies in 2030 and 2050

5. Conclusion

In this study, the transition to sustainable aviation fuels is explored. The synergy among renewable energy, direct air capture for CO₂, and power-to-liquid technologies provides a potential clean solution to power the air transport industry of the future. However, several challenges have been identified through this investigation. First, to transition to SAF thru DAC and RE, additional renewable energy capacity is expected. In the case of the Philippines, it would need to increase its target RE goals by 50% by 2050. Next, land use requirements for the expected energy needs for this transition is huge. Alternative technologies such as offshore wind, hydropower, and nuclear may provide preferred results. Lastly, the calculated CAPEX for this transition is 7.9 billion euros for the study area. The high CAPEX would ultimately determine the cost of e-fuel. Decreased technology prices would greatly benefit the economics of this technology.

6. References

Academic References

- Rojas-Michaga, M. F., Michailos, S., Cardozo, E., Akram, M., Hughes, K. J., Ingham, D., & Pourkashanian, M. (2023). Sustainable aviation fuel (SAF) production through power-to-liquid (PtL): A combined techno-economic and life cycle assessment. *Energy Conversion and Management*, 292. <https://doi.org/10.1016/j.enconman.2023.117427>
- Grim, R. G., Ravikumar, D., Tan, E. C. D., Huang, Z., Ferrell, J. R., Resch, M., Li, Z., Mevawala, C., Phillips, S. D., Snowden-Swan, L., Tao, L., & Schaidle, J. A. (2022). Electrifying the production of sustainable aviation fuel: the risks, economics, and environmental benefits of emerging pathways including CO₂. *Energy and Environmental Science*, 15(11), 4798–4812. <https://doi.org/10.1039/d2ee02439j>
- Bolinger, M., & Bolinger, G. (2022). Land Requirements for Utility-Scale PV: An Empirical Update on Power and Energy Density. *IEEE Journal of Photovoltaics*, 12(2), 589–594. <https://doi.org/10.1109/JPHOTOV.2021.3136805>
- Seymour, K., Held, M., Stolz, B., Georges, G., & Boulouchos, K. (2024). Future costs of power-to-liquid sustainable aviation fuels produced from hybrid solar PV-wind plants in Europe. *Sustainable Energy and Fuels*, 8(4), 811–825. <https://doi.org/10.1039/d3se00978e>
- Sens, L., Neuling, U., & Kaltschmitt, M. (2022). Capital expenditure and levelized cost of electricity of photovoltaic plants and wind turbines – Development by 2050. *Renewable Energy*, 185, 525–537. <https://doi.org/10.1016/j.renene.2021.12.042>
- Bourgeois, L., Ahmann, M., Albertos, F., Allevi, B. C., Bergh, M., Cabrera, M., Csoka, G. A., de Montessus, H., Delebecque, J.-C., Edwards, N., Haupt-Herting, S., Herlakian, N., Hernandez, S., Holst, B. A., Kozakowski, G., Krupa, P., Lobban, J., Loonen, R., Mackenzie, A. R. D., ... Pokela, S. (2012). *EU refinery energy systems and efficiency Prepared for the CONCAWE Refinery Management Group by its Special Task Force RT/STF-1*. www.concawe.org
- Lilley, W., Schmidt, P., Weindorf, W., Raksha, T., Failer, S., Astono, Y., & Houghton, T. (n.d.). *E-Fuels: A techno-economic assessment of European domestic production and imports towards 2050-Update*.

Online References

https://www.theglobaleconomy.com/rankings/jet_fuel_consumption/

<https://www.wssenergy.com/post/is-e-fuel-production-a-responsible-way-to-use-renewable-energy>

<https://www.withersworldwide.com/en-gb/insight/read/key-considerations-for-renewable-energy-in-vestments-in-the-philippines>

<https://aboitizpower.com/news/power-generation/powering-the-philippine-economy-electricity-generation-in-economic-development>

<https://energytracker.asia/renewable-energy-in-the-philippines-current-state-and-future-roadmap/>

<https://www.eerenewables.co.uk/solar-guides/how-many-solar-panels-needed-to-generate-1-megawatt/>

<https://docs.nrel.gov/docs/fy09osti/45834.pdf>